

STA 35C – Homework 3

Submission due: Thu, October 15 at 11:59 PM PT

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Instructions: Upload a single PDF file to Gradescope via Canvas (“Homework 3” under “Assignments”). Name the file using the prefix of your UC Davis email ID and the homework number (e.g., `dgsong_hw3.pdf`). Include “STA 35C,” your name, and the last four digits of your student ID on the front page. For coding problems, include your R code, relevant output, and any requested plots/tables in the PDF. No late submissions will be accepted; any submission received after the deadline will receive 0 points. For full information about submission requirements and the late submission policy, see the syllabus.

Problem 1 (25 points in total + 5 bonus points).

You are analyzing an outcome measure Y (e.g., blood pressure or some clinical score) collected from n patients. Some patients received a *new treatment* ($D = 1$), while others received *standard* (control) treatment ($D = 0$). You also have an additional continuous predictor X (e.g., patient age or baseline risk). You suspect that Y may differ substantially between the two groups, possibly due to the different treatments introduced. Throughout, we assume that *smaller values of Y are better*.

Your dataset consists of n patient records, given as:

```
patients <- data.frame(
  Y = c(-9.3704, -9.2399, -8.1464, -7.1728, -6.3464,
        -5.8743, -5.6763, -1.8406, -0.8549, 1.0222),
  D = c(1, 1, 0, 1, 1, 0, 1, 0, 0, 0),
  X = c(39.107, 38.245, 24.318, 36.825, 29.314,
        23.497, 31.137, 14.919, 13.691, 10.631)
)
```

In this problem, you will compare the two groups using methods learned so far. Interpret the group comparisons as associations, not established causal effects. For each t -test below, use the usual model-based standard errors under the fitted model’s working assumption of independent, mean-zero Gaussian errors with constant variance, conditional on its predictors. You may use R throughout.

(a) (5 points) Initially, compare the average outcome in each group:

$$\bar{y}_{\text{ctrl}} = \frac{1}{n_0} \sum_{i: d_i=0} y_i, \quad \bar{y}_{\text{treat}} = \frac{1}{n_1} \sum_{i: d_i=1} y_i,$$

where n_0 is the number of patients with $D = 0$, and n_1 the number with $D = 1$ (here $n_0 = n_1 = 5$).

Compute the sample mean of Y in each group and the raw difference $\bar{y}_{\text{treat}} - \bar{y}_{\text{ctrl}}$. Based only on these means, which group has lower observed outcomes?

- (b) (7 points) Now consider a simple linear model that only takes D into account, *ignoring* X . Fit $Y = \beta_0 + \beta_1 D + \varepsilon$. Report and interpret $\hat{\beta}_0$ and $\hat{\beta}_1$, and verify that $\hat{\beta}_1$ equals the raw difference in part (a). Test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$ at the 5% level; report the p -value and a conclusion in context.

- (c) (7 points) Next, you include the additional predictor X to obtain an adjusted model:

$$Y = \beta_0 + \beta_1 D + \beta_2 X + \varepsilon.$$

Fit this model, report and interpret $\hat{\beta}_1$ and $\hat{\beta}_2$, and compare the adjusted group difference with part (b). Check the observed range of X in each group: does their overlap, or lack of overlap, matter for this comparison and why?

- (d) (6 points) Finally, fit the model with an interaction term

$$Y = \beta_0 + \beta_1 D + \beta_2 X + \beta_3 (D \times X) + \varepsilon$$

which is equivalent to

$$Y = \begin{cases} \beta_0 + \beta_1 + (\beta_2 + \beta_3)X + \varepsilon & \text{if } D = 1, \\ \beta_0 + \beta_2 X + \varepsilon & \text{if } D = 0. \end{cases}$$

Write the fitted mean-response line for each group and interpret $\hat{\beta}_3$. Test $H_0 : \beta_3 = 0$ against $H_1 : \beta_3 \neq 0$ at the 5% level, reporting its p -value. Is there evidence of different slopes? Distinguish failure to reject equal slopes from establishing equal slopes.

- (e*) (5 bonus points) Using the interaction model from part (d), estimate the average fitted treatment–control contrast at the same observed predictor values:

$$\hat{\tau} = \frac{1}{10} \sum_{i=1}^{10} [\hat{\mathbb{E}}(Y \mid D = 1, X = x_i) - \hat{\mathbb{E}}(Y \mid D = 0, X = x_i)].$$

Express $\hat{\tau}$ in terms of the fitted coefficients and $\bar{x}$, calculate its value, and interpret its sign without making a causal claim.

Problem 2 (25 points in total).

A company administers a mandatory test to new hires in a training program. Each candidate either *Passes* ($Y = 1$) or *Fails* ($Y = 0$). The HR department believes that prior work experience (X , in months) might influence the probability of passing. Treat $Y = 1$ as the positive class throughout.

- (a) (5 points) Assume the logistic model

$$p(x) = \Pr(Y = 1 \mid X = x) = \frac{1}{1 + e^{-(\alpha + \beta x)}}.$$

Interpret α and β in terms of log-odds, and give the multiplicative change in the odds of passing for a one-month increase in experience.

- (b) (7 points) Suppose the fitted coefficients are $\hat{\alpha} = -3.0$ and $\hat{\beta} = 0.07$. HR assigns candidates to an “advanced” track if $\hat{p}(x) \geq 0.4$ and a “basic” track otherwise.

- (i) Compute $\hat{p}(x)$ for $x = 20$ months and predict the assigned track.
- (ii) If a candidate actually passes ($Y = 1$) but is assigned to the “basic” track under the 0.4 cutoff, is this a false positive or a false negative? Briefly explain.

- (iii) Discuss the pros and cons of using 0.4 rather than 0.5 as a threshold. In particular, consider which type of misclassification might be costlier for the company.
- (c) (8 points)** Implement the following in an R script and report your results.
- (i) Using `set.seed(2026)`, simulate a dataset of size $n = 50$ by letting $X \sim \text{Uniform}(0, 30)$ and
- $$\Pr(Y = 1 \mid X = x) = \frac{1}{1 + e^{-(-5 + 0.25 x)}}.$$
- Generate Y accordingly (*Hint*: Use `rbinom()` in R).
- (ii) Fit a logistic model in R (via `glm(..., family=binomial)`), report the estimated coefficients, and briefly compare them with ($\alpha = -5$, $\beta = 0.25$).
- (iii) Plot the fitted logistic curve on a grid of x -values from 0 to 30, together with the observed data points.
- (d) (5 points)** Use the same simulated data from part (c).
- (i) Fit Y on X by ordinary least squares. On the same grid used in part (c), add its fitted line and the true probability curve to the plot. Label all three curves and adjust the vertical axis as needed to show predictions outside $[0, 1]$.
- (ii) For each fitted model, do all predictions on this grid lie in $[0, 1]$? Is that sufficient for accurate probability estimates? Answer in 1–2 sentences.

Problem 3 (25 points in total).

Consider the logistic model for binary classification:

$$\log \frac{p(X)}{1 - p(X)} = \beta_0 + \beta_1 X_1 + \beta_2 X_2, \quad \text{where} \quad p(X) = \Pr[Y = 1 \mid X].$$

Given $X = (x_1, x_2)$, classify $\hat{Y} = 1$ if $\hat{p}(x_1, x_2) \geq p^*$ and $\hat{Y} = 0$ otherwise, where $\hat{p}$ uses the fitted coefficients. For parts (a)–(b), suppose the estimated coefficients are:

$$\hat{\beta}_0 = -1.2, \quad \hat{\beta}_1 = 2.0, \quad \hat{\beta}_2 = -0.5.$$

- (a) (6 points)** You observe a new test point $x_{\text{test}} = (x_1, x_2) = (2, 3)$. Decide whether $\hat{y}_{\text{test}} = 1$ or $\hat{y}_{\text{test}} = 0$ at $p^* = 0.5$.
- (b) (6 points)** Derive the equation of the decision boundary for $p^* = 0.5$, and sketch this line in the (x_1, x_2) plane, indicating where $\hat{Y} = 1$ vs. $\hat{Y} = 0$.
- (c) (13 points total)** Using R, fit the logistic model `Label ~ X1 + X2` on the following dataset of 100 points. For parts (i) and (ii), use the fitted probabilities from this model on the same dataset, classify $\hat{Y} = 1$ when $\hat{p} \geq p^*$, and treat $Y = 1$ as the positive class.

```
set.seed(1) # for reproducibility
X1_1 <- rnorm(40, mean = 2, sd = 1)
X2_1 <- rnorm(40, mean = 2, sd = 1)

# Generate (X1, X2) for label=0
X1_0 <- rnorm(60, mean = 0, sd = 1.2)
X2_0 <- rnorm(60, mean = 0, sd = 1.2)
```

```
# Combine into a single dataset
X1 <- c(X1_1, X1_0)
X2 <- c(X2_1, X2_0)
Y <- c(rep(1, 40), rep(0, 60))

# Create a data frame
df <- data.frame(X1, X2, Label = Y)
```

- (i) **(6 points)** Create a confusion matrix, and compute the true positive rate (TPR = sensitivity), and the false positive rate (FPR = 1 - specificity) at $p^* = 0.5$.
- (ii) **(7 points)** For $p^* \in \{0, 0.05, 0.1, \dots, 0.95, 1\}$, compute TPR and FPR, then plot the resulting approximate ROC curve (FPR, TPR), connecting points in threshold order. Use axis limits $[0, 1]$ on both axes. In 2–3 sentences, explain how lowering the threshold affects false positives and false negatives, and describe one practical scenario where this cost tradeoff can be significant.

Problem 4 (25 points total + 5 bonus points).

You catch crabs of two species, A and B, recording these weights (pounds):

Species A: $x \in \{1, 2, 3\}$, Species B: $x \in \{3, 4, 5, 6\}$.

Assume Gaussian class-conditional distributions with means μ_A, μ_B and a common variance σ^2 . For parts (a)–(d), use the following sample-proportion priors: $\hat{\pi}_A = \frac{3}{7}$, $\hat{\pi}_B = \frac{4}{7}$.

- (a) **(5 points)** Compute the sample means $\bar{x}_A, \bar{x}_B$ and the pooled sample variance

$$s^2 = \frac{1}{n_A + n_B - 2} \left(\sum_{i \in A} (x_i - \bar{x}_A)^2 + \sum_{i \in B} (x_i - \bar{x}_B)^2 \right).$$

- (b) **(5 points)** Write the linear discriminant functions $\delta_A(x)$ and $\delta_B(x)$, using $\bar{x}_A, \bar{x}_B, s^2$ obtained above.
- (c) **(5 points)** If $x_{\text{new}} = 3.2$, which species do you predict? Show your reasoning.
- (d) **(5 points)** Suppose missing a crab of Species B is especially costly. Thus, you decided to predict A if the fitted posterior probability $\widehat{\Pr}(Y = A \mid X = x) \geq p^*$, and predict B otherwise, where $p^* \in (0.5, 1)$ (e.g. $p^* = 0.9$). Express this rule in terms of $\delta_A(x) - \delta_B(x)$, state the decision boundary as a function of p^* , and apply it to $x_{\text{new}} = 3.2$.
- (e) **(5 points)** To isolate the effect of the priors, update only the class priors to the sample proportions that would result from adding three Species A observations, keeping the estimates of $\bar{x}_A, \bar{x}_B, s^2$ fixed at their values from part (a). Using $p^* = 0.5$, would your prediction at $x_{\text{new}} = 3.2$ change from part (c)?
- (f*) **(5 bonus points)** Suppose that Species A and B are not Gaussian-distributed but Laplace-distributed with class means μ_A, μ_B , a *common* scale parameter b , and the original class priors $\pi_A = 3/7$ and $\pi_B = 4/7$. The Laplace probability density function with mean μ and variance $2b^2$ is

$$f(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right).$$

- (i) Draw the graph of $f(x; 0, 1)$ and compare it with the graph of Gaussian density $\mathcal{N}(0, 2)$, which has the same mean and variance.
- (ii) Derive the discriminant function under Laplace density and compare it with the linear discriminant function under Gaussian. (Hint: consider $\log(\pi_k f_k(x))$.)